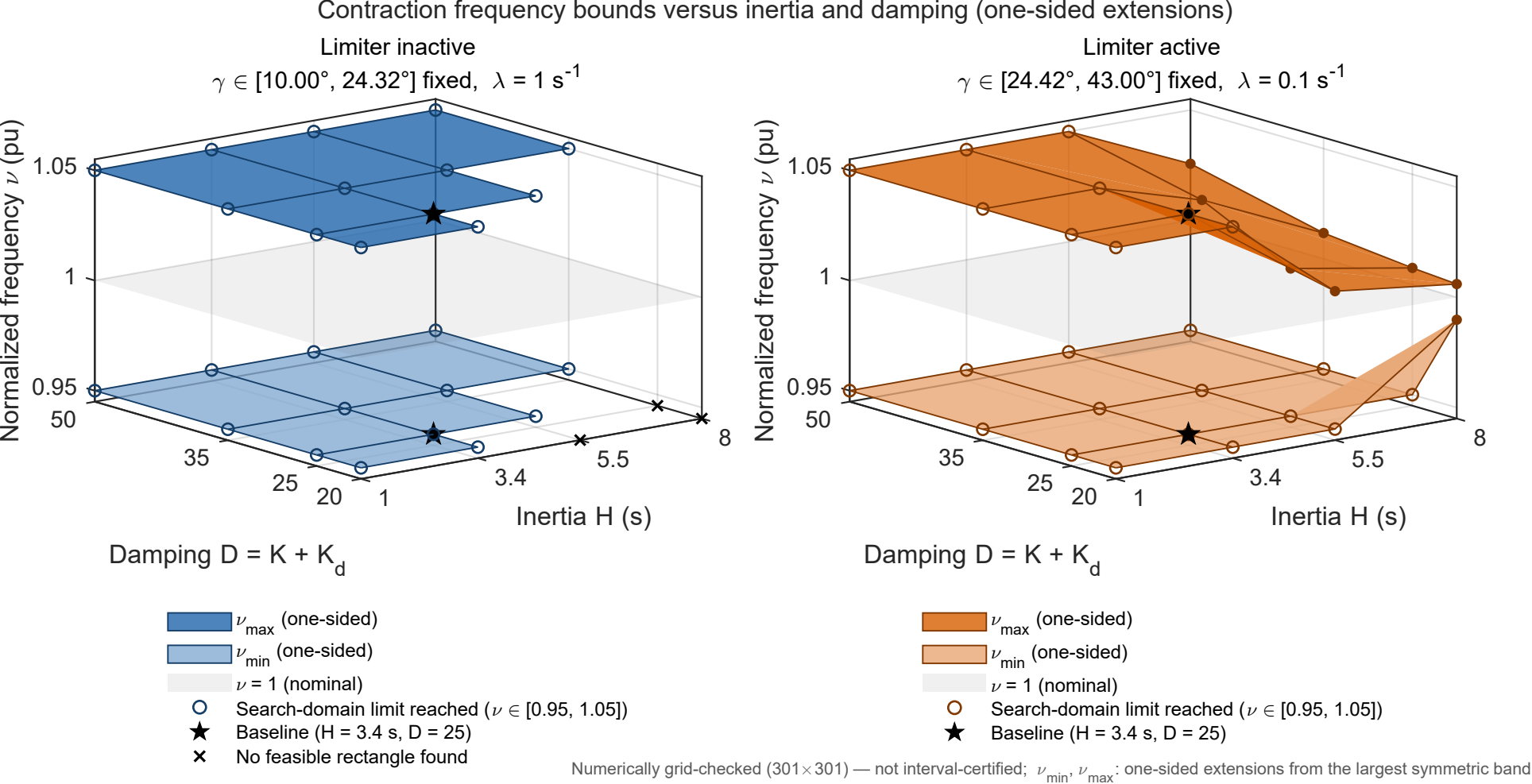
Contraction frequency bounds versus inertia and damping (one-sided extensions)
Limiter inactive
$\gamma \in [10.00°, 24.32°]$ fixed, $\lambda = 1\ s^{-1}$
Limiter active
$\gamma \in [24.42°, 43.00°]$ fixed, $\lambda = 0.1\ s^{-1}$
Normalized frequency $\nu$ (pu)
1.05
1
0.95
Inertia H (s)
1
3.4
5.5
8
Damping D = K + K$_d$
50
35
25
20
$\nu_{max}$ (one-sided)
$\nu_{min}$ (one-sided)
$\nu$ = 1 (nominal)
Search-domain limit reached ($\nu \in [0.95, 1.05]$)
Baseline (H = 3.4 s, D = 25)
No feasible rectangle found
Numerically grid-checked (301×301) — not interval-certified; $\nu_{min}$, $\nu_{max}$: one-sided extensions from the largest symmetric band